

Empirical Industrial Organization, Spring 2026

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Homework 4: Demand, Costs, and Fixed-Cost Bounds in an Airline Market

Instructions

- Submit your homework by July 3rd at 8pm.
- Your submission must include: (i) the console output of the scripts, and (ii) a short report with the requested tables and your written interpretation.
- Send the homework to `schacon@fen.uchile.cl` and `jpazm@fen.uchile.cl`.
- You may work in groups of two; each group submits one report.
- Use Python with `pyblp`, `numpy`, `pandas`, and `matplotlib`. You are given the data, the scripts, and a data dictionary.
- Enjoy!

Overview

You will run a complete structural pipeline on a (simulated) international airline market and explain what each stage recovers. A “flight” (product) is a carrier’s nonstop service on a route; a “market” is a route (origin–destination) $\times$ period; carriers AA and BB are a pair that could merge. The pipeline climbs the usual ladder: estimate *demand* (how travelers substitute across flights), recover *marginal costs* from the pricing first-order conditions, and then *bound fixed costs* — the cost of keeping a route on the schedule at all — by revealed preference (Eizenberg, 2014). Fixed costs are never observed, so the data only *bound* them.

Parts 1–2 (required) are run-and-interpret: you execute the provided scripts and explain the output. Part 3 (optional) uses the fixed-cost bounds to ask what a merger between *A* and *B* does to travelers when the merged carrier may cancel overlapping routes, and how the answer moves with the degree of coordination κ . All derivations are in the Appendix.

1 Part 1 — Run the pipeline

Run the three scripts *in order*; each saves an object the next one needs. The third resamples shocks and re-solves equilibria, so it takes a few minutes.

- (a) `run_demand_airlines.py` $\rightarrow$ `informacion_demanda.obj`. Estimates the random-coefficients (BLP) airline demand. Confirm it converged and record the estimated coefficients (fare, frequency, premium, service; the random-coefficient σ ’s; the income interactions π).

- (b) `run_supply_airlines.py` → `informacion_oferta.obj`. Assumes pre-merger multiproduct Bertrand-Nash and inverts the pricing first-order conditions to recover each flight's marginal cost mc_{jt} and cost shock ω_{jt} . Record the cost-equation coefficients γ and the share of flights whose mc had to be floored.
- (c) `run_fixedcosts_airlines.py` → fixed-cost bounds and a fitted fixed-cost function (Eizenberg). Record how many markets solved, the median upper bound (offered routes) and median lower bound (non-offered routes), and the fitted coefficients of $F_j = \phi'W_j$.

Deliverables. One table reproducing each script's headline numbers (demand coefficients; cost coefficients and floored share; bound medians and the fixed-cost function). No interpretation yet — that is Part 2.

2 Part 2 — Interpret what you computed

Demand

- (a) Report the sign and magnitude of the fare coefficient, the mean own-fare elasticity, and the mean Lerner markup. Explain in two sentences why plain logit gets substitution wrong, and why that matters for any merger analysis.
- (b) `run_demand` also prints, for every route where A and B both operate, the diversion ratio between their two flights — $D_{jk} = -(\partial s_k / \partial p_j) / (\partial s_j / \partial p_k)$, the fraction of travelers who leave flight j when its fare rises and switch to k — in both directions and as the symmetrized $\bar{D}_{jk} = \frac{1}{2}(D_{jk} + D_{kj})$. Report this table of overlapping routes with their diversion, and explain what a diversion ratio measures. Since each such route pits one A flight against one B flight, $\bar{D}_{AB}$ is simply how close the two substitutes are — which is what predicts the withdrawal pressure on that route in Part 3.

Costs

- (c) Interpret the cost equation: which cost shifters and characteristics raise marginal cost, and by how much? Report the floored- mc share and state what a *large* share would mean — a recovered markup above the fare rejects the demand/conduct specification (the Corts critique).

Fixed-cost bounds

- (d) Explain why an *offered* route delivers an *upper* bound on its fixed cost, while a *non-offered* route delivers a *lower* bound. (The carrier flew the route only if expected variable profit covered the fixed cost; it stayed out only if it did not.) Why do the two sets of bounds not cross?
- (e) Read the fitted fixed-cost function $F_j = \phi'W_j$: which route characteristics raise the cost of offering service? Interpret the premium split in the output (do premium routes cost more to operate?). One paragraph.

Deliverables. The A – B diversion table (all overlapping routes), and short written answers to (a)–(e).

3 Part 3 — A merger with route withdrawal

It reuses everything above: **the upper bounds from Part 2 are exactly the fixed-cost ceilings that decide which routes survive a merger.** Carriers A and B merge into M , jointly own their flights, and may cancel overlapping ones; the parameter $\kappa \in [0, 1]$ sets how strongly they coordinate fares ($\kappa = 1$ full integration, $\kappa = 0$ joint ownership but no fare coordination). Use the provided `merger_engine.py` (its `delta_vp` and `consumer_surplus` functions); running it as a script works a full route end to end.

- (a) **Understand it on one route** ($\kappa = 0.5$). Pick any head-to-head route from your Part 2(b) table, set `ROUTE` in `merger_engine.py`, and run it. For each of A 's and B 's flights it reports the pre-merger incremental profit ΔVP_j^{pre} (the fixed-cost upper bound) and the post-merger value $\Delta VP_j^{\text{post}}(\kappa)$ under joint ownership $\Omega(\kappa)$; a flight is *at risk* when $\Delta VP_j^{\text{post}} < \Delta VP_j^{\text{pre}}$ — the merger pushed its value below its fixed-cost ceiling, so it *could* be cancelled. Treat the at-risk set as the pool of *candidates*, not the flights actually dropped: which of them (if any) is cut depends on the unobserved fixed costs, and the engine's consistency check is what turns "both at risk" into a single actual cancellation. It then prints the consumer-surplus interval — best case (no cancellation) vs. the consumer-worst *consistent* cancellation — with the repricing/withdrawal decomposition and width. Read these off and explain them for your route. (Note the interaction the engine checks: once one redundant flight is cancelled, its sibling recaptures the traffic and becomes too valuable to cancel, so the worst case is a single cancellation, not route exit.)
- (b) **Sweep all overlapping routes and all κ .** Loop the engine's `delta_vp` and `consumer_surplus` over *every* head-to-head market (every market where A and B both operate) and over $\kappa \in \{0, 0.25, 0.5, 0.75, 1\}$. For each (route, κ) record the at-risk set and the ΔCS interval — its two endpoints, width, and repricing/withdrawal split. *This is a few minutes of compute; debug with a small n_{draws} (e.g. 4), then raise it.*
- (c) **Aggregate and interpret.** For each κ , summarize across routes — e.g. the median interval width and the median withdrawal effect — and plot them against κ . Then connect back to Part 2(b): do routes with higher symmetrized diversion $\bar{D}_{AB}$ show larger withdrawal effects and wider intervals? For the network as a whole, discuss how the harm splits between fare coordination (repricing) and portfolio rationalization (withdrawal), and what each does as κ moves from 0 to 1.

Deliverables. For one route: the per-flight ΔVP table and the ΔCS interval with its decomposition. For the full sweep: a figure of the median ΔCS interval (both endpoints) against κ , a scatter of interval width against the Part 2(b) diversion across routes, and a short written interpretation.

A Pricing equilibrium and the conduct assumption

In market t with flights $\mathcal{J}$, fares p , shares $s(p)$, and marginal costs mc , let $\Delta(p)$ be the matrix with $\Delta_{jk} = \partial s_k / \partial p_j$. An ownership matrix Ω has $\Omega_{jk} \in [0, 1]$ equal to the weight flight j 's owner places on flight k 's profit. Bertrand fare-setting solves

$$s(p) + (\Omega \odot \Delta(p))(p - mc) = 0 \iff p = mc - (\Omega \odot \Delta(p))^{-1} s(p), \quad (1)$$

with $\odot$ the Hadamard product. **Pre-merger** Ω^{pre} sets $\Omega_{jk} = 1$ within a carrier and 0 across carriers; inverting (1) under it gives the marginal costs of Part 1(b). **Post-merger** $\Omega(\kappa)$ equals Ω^{pre} except that the A – B cross block is κ (built in the engine with `build_ownership`). Consumer surplus depends only on fares and demand, so it is computed at the equilibrium fares under whichever Ω applies.

B Fixed-cost bounds and the merger keep rule

Timing (Eizenberg, 2014). A carrier picks its route portfolio knowing only the *distribution* of the shocks (ζ, ω) , then sets fares once they realize. Portfolio choices therefore compare *expected* variable profits, with no selection on realized shocks — hence the expectations below are integrated by resampling (ζ, ω) with common random numbers (what `run_fixedcosts` does).

Variable profit and bounds. For a flight set S and ownership Ω , let $p^*(S)$ solve (1); carrier f 's variable profit is $\text{VP}_f(S) = \sum_{j \in S \cap \mathcal{J}_f} (p_j^* - mc_j) s_j M_t$. The incremental value of j is the leave-one-out difference. If j was offered, the carrier preferred keeping it, so $F_j \leq \Delta \text{VP}_j^{\text{pre}}$ (an *upper* bound); if a feasible route k was *not* offered, adding it was not worthwhile, so $F_k \geq \Delta \text{VP}_k$ (a *lower* bound). Selection runs in opposite directions for the two groups, which is why the bounds do not cross.

Merger keep rule and the interval. Post-merger the carrier keeps j iff $F_j \leq \Delta \text{VP}_j^{\text{post}}(\kappa)$, with $\Delta \text{VP}_j^{\text{post}} = \mathbb{E}[\text{VP}_M(\mathcal{J}) - \text{VP}_M(\mathcal{J} \setminus \{j\})]$ and $\text{VP}_M = \text{VP}_A + \text{VP}_B$. Since F_j is only known to lie in $[0, \Delta \text{VP}_j^{\text{pre}}]$, flight j is *at risk* iff $\Delta \text{VP}_j^{\text{post}} < \Delta \text{VP}_j^{\text{pre}}$ (the merger lowered its value because it now recaptures traffic from a co-owned sibling; the gap is largest for high-diversion routes). The cancelled set is only set-identified, so the merger's effect on consumer surplus is an interval: the best case cancels nothing, and the worst case is the consumer-worst cancellation that is still revealed-preference-consistent. Consistency matters: cancelling a flight raises its sibling's value (no more internal cannibalization), so cancelling several redundant flights at once is generally *not* optimal — the engine checks this and takes the worst single cancellation rather than route exit. The interval width is the harm the data cannot resolve, because it hinges on the unobserved fixed costs.

Reference

Eizenberg, Alon. 2014. "Upstream Innovation and Product Variety in the U.S. Home PC Market." *Review of Economic Studies* 81(3): 1003–1045.